Jingtian Ji

jijingtian@ttic.edu

lucienji.github.io • Google Scholar • github.com/LucienJi

Education

Toyota Technological Institute at Chicago (TTIC) • Chicago, IL, USA — Sep. 2024 – Present
Ph.D. Student, Computer Science (Robotics) • Advisor: Prof. Matthew R. Walter

Shanghai Jiao Tong University • Shanghai, China — Sep. 2022 – Mar. 2024
M. Eng. Electronic and Information Engineering • GPA: 3.96/4.00

École Polytechnique • Palaiseau, France — Mar. 2020 – Sep. 2022
Diplôme d'Ingénieur (M. Eng.) in Computer Science • GPA: 3.81/4.00

Shanghai Jiao Tong University • Shanghai, China — Sep. 2017 – Mar. 2020
B. Eng. Information Engineering • GPA: 3.93/4.30 • Rank: 1/12

Jingcai High School • Shanghai, China — Sep. 2015 – Jun. 2017

Publications

* Equal contribution.

- **Jingtian Ji**, Xuefeng Liu, Matthew R. Walter, "Compose Your Oracles: Off Policy Improvement with Aggregated Guidance," *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. Code.
- Haoran Chen*, **Jingtian Ji***, Samuel Wheeler, Kaylene Caswell Stocking, Matthew R. Walter, "Beyond Reconstruction: What Matters in Action Tokenization for Robot Policies?" *Under review at the International Conference on Learning Representations (ICLR)*, 2027. arXiv:2610.09170.
- Haoran Chen*, **Jingtian Ji***, Samuel Wheeler, Kaylene Caswell Stocking, Matthew R. Walter, "CAP: Codebook-Aligned Prediction for Tokenized Robot Policies," *Under review at the IEEE International Conference on Robotics and Automation (ICRA)*, 2027. arXiv:2610.09178.
- Luzhe Sun*, **Jingtian Ji***, Haoran Chen, Jiawei Zhou, Matthew R. Walter, "Flow-based Policy Adaptation without Policy Updates," *arXiv preprint* arXiv:2606.06461, 2026.
- Tianchong Jiang, **Jingtian Ji**, Xiangshan Tan, Jiading Fang, Anand Bhattad, Vitor Guizilini, Matthew R. Walter, "Do You Know Where Your Camera Is? View-Invariant Policy Learning with Camera Conditioning," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. **Best Paper Award on Robot Learning.** arXiv:2510.02268.
- Xiangshan Tan, **Jingtian Ji**, Tianchong Jiang, Pedro Lopes, Matthew R. Walter, "HapCompass: A Rotational Haptic Device for Contact-Rich Robotic Teleoperation," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. arXiv:2603.30042.
- Luzhe Sun, **Jingtian Ji**, Xiangshan Tan, Matthew R. Walter, "FlashBack: Consistency Model–Accelerated Shared Autonomy," *Conference on Robot Learning (CoRL)*, 2025. arXiv:2505.16892.
- Aakaash Narayanan, Jose Berlioz, Kit Danison-Fieldhouse, Kyle Hazelwood, Maira Khan, Jason St. John, Nhan Tran, Andrew Whitbeck, **Jingtian Ji**, Matthew R. Walter, "Surrogate Model for Third-Integer Resonance Extraction at the Fermilab Delivery Ring," *North American Particle Accelerator Conference (NAPAC)*, 2025. doi:10.18429/JACoW-NAPAC2025-MOP089.
- Andrew Whitbeck, Jose Berlioz, Kit Danison-Fieldhouse, Kyle Hazelwood, Maira Khan, Jovan Mitrevski, Aakaash Narayanan, Jason St. John, Nhan Tran, **Jingtian Ji**, Matthew R. Walter, "Fast Adaptive Neural Control of Resonant Extraction at Fermilab," *North American Particle Accelerator Conference (NAPAC)*, 2025. arXiv:2511.19099.
- Jianming Ma, **Jingtian Ji**, Yue Gao, "Adversarial Constrained Policy Optimization: Improving Constrained Reinforcement Learning by Adapting Budgets," *arXiv preprint* arXiv:2410.20786, 2024.
- Buqing Nie, **Jingtian Ji**, Yangqing Fu, Yue Gao, "Improve Robustness of Reinforcement Learning against Observation Perturbations via ℓ_∞ Lipschitz Policy Networks," *Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI-24)*, vol. 38(13), pp. 14457–14465, 2024.
- **Jingtian Ji**, Buqing Nie, Yue Gao, "DAGA: Dynamics-Aware Reinforcement Learning with Graph-Based Rapid Adaptation," *IEEE Robotics and Automation Letters*, 8(4):2189–2196, 2023.

Work Experience

Game AI Algorithm Intern • NetEase AI Lab — Apr. 2023 – Aug. 2023

- Developed a Game AI training framework based on curriculum learning and self-play algorithms.
- Implemented a distributed reinforcement learning framework using CPU clusters and multi-GPUs.
- Employed Elo score to identify the optimal diverse and effective models; maintained a long-term model pool to enhance self-play performance.

- Achieved 1st place in the Olympic competition at IJCAI.

Service & Awards

- **Best Paper Award on Robot Learning**, ICRA 2026 (for "Do You Know Where Your Camera Is?").
- Teaching Assistant, *Artificial Intelligence: Theory and Applications*, 2023: designed a reinforcement learning project, held interactive Q&A sessions, and gave feedback on assignments.
- China Scholarship Council (CSC) Scholarship, 2020–2022.
- Academic Excellence Scholarship, 2019.

Skills

- **Languages:** Mandarin (native); French (DELF B2); English (TOEFL 110, GRE 163V/170Q/4.0AW; extensive scientific writing experience).
- **Programming:** Python, C++, Java.